

Week 5 · Tuesday — Score Real Work, Many Ways

TCE 318P FA26 (Edwards)

In class · assessment studio. The task, four real (anonymous) grade-4 responses, and the rubrics your table will score with.

The task

An amusement park has games, rides, and shows. The total number of games, rides, and shows is 70. There are 34 rides. There are two times as many games as shows.

How many games are there? _____ How many shows are there? _____

Use numbers, words, or drawings to show how you got your answer.

The rubrics

Rubric 1 *Holistic · built for this task · one overall level*

Level	What earns it
Extended	24 games and 12 shows, with correct explanation or work.
Satisfactory	Subtraction error but games and shows in the correct 2 : 1 ratio; OR 12 games and 24 shows with work; OR 24 games and 12 shows with no work.
Partial	Finds 36 and a 2 : 1 ratio (but not 24 : 12) with a sum under 36; OR 36 and 18; OR 72 and 36; OR a process that shows understanding but not the correct ratio.
Minimal	Finds 36 by subtracting (or adding up to 70); OR games + shows = 36; OR a 2 : 1 ratio but nothing else correct.
Incorrect	Incorrect response.

Rubric 2 *Holistic · general · one dimension only — mathematical communication*

Score	Uses mathematical language that is...
4	Uses mathematical language (terms, symbols, representations) that is highly effective, accurate, and thorough, to describe operations, concepts, and processes.
3	...partially effective, accurate, and thorough.
2	...minimally effective and accurate.
1	An incorrect response — an attempt is made.
0	Off task, off topic, illegible, blank, or insufficient to score.

Task and work samples: NAEP grade-4 released item 2011-4M8 (public domain). Rubric 3: Charles, Lester & O'Daffer (1987), as shown in McGatha & Darcy (2010), Fig. 3. Rubrics 2 & 4: Temple ISD and Maryland SDE scoring scales. No student names appear here.

Rubric 3 *Holistic · general · the “did they get it right?” lens*

Score	Description
3	Response is exemplary, detailed and clear.
2	Response is generally correct.
1	Response is partially correct, but lacks clarity.
0	No response, or response is incorrect.

Rubric 4 *Analytic · general · a separate score for each part of problem solving***Understanding the problem**

Pts	
6	Complete understanding of the problem.
3	Part of the problem misunderstood or misinterpreted.
0	Complete misunderstanding of the problem.

Planning a solution

Pts	
6	Plan could have led to a correct solution if implemented properly.
3	Partially correct plan based on correct interpretation of part of the problem.
0	No attempt, or totally inappropriate plan.

Getting an answer

Pts	
3	Correct answer and correct label for the answer.
2	Incorrect answer that follows logically from an inappropriate plan.
1	Copying error, computational error, or partial answer (problem with multiple answers).
0	No answer, or wrong answer based on an inappropriate plan.

The work samples

Four real, anonymous grade-4 responses. Use your group's assigned rubric to score each work sample. We'll share findings collaboratively during our debrief.

Student Work Sample	Scoring
How many games are there? <u>24</u> How many shows are there? <u>12</u> $\begin{array}{r} 6 ^{10} \\ 7 \times \\ - 34 \\ \hline 36 \end{array}$ $\begin{array}{r} 12 \\ 3 \overline{)36} \\ \hline \end{array}$ $\begin{array}{r} 12 \\ \times 2 \\ \hline 24 \end{array}$	
How many games are there? <u>12</u> How many shows are there? <u>24</u> You do $70 - 34 = 36$. The you estimate what number and 2 times as much as the number plus 34 equals 70. So I got $12 + 24 + 34 = 70$.	
How many games are there? <u>24</u> How many shows are there? <u>12</u> $70 - 34 = 36$ so $36 \div 3 = 12$ that's the number of shows there are then you subtract 34.	
How many games are there? <u>24</u> How many shows are there? <u>12</u>	